

Artificial Intelligence & Machine Learning — Task 2

Feature Engineering, Model Optimization & Performance Comparison
Submitted by: Abhishek Pal

1. Objective

The objective of this task was to build an enhanced House Price Prediction system using the California Housing Dataset. The focus was on applying real-world Machine Learning practices: feature scaling, training multiple regression models, evaluating them using standard metrics, and selecting the best-performing model with proper justification.

2. Methodology

Dataset: The California Housing Dataset was loaded via scikit-learn. It contains 8 input features including Median Income, House Age, Average Rooms, and geographic coordinates, with Median House Value as the target variable.

Feature Scaling: StandardScaler was applied to all input features before model training. This ensures features with different numeric ranges contribute equally to the model, preventing dominant features from skewing results.

Train-Test Split: The dataset was split into 80% training and 20% testing sets using random_state=42 for reproducibility. Models were evaluated exclusively on unseen test data to avoid data leakage.

Model Training: Three regression algorithms were trained and compared: (1) Linear Regression — used as the baseline model, (2) Ridge Regression (alpha=1.0) — to reduce overfitting via L2 regularization, and (3) Decision Tree Regressor (max_depth=5) — to capture non-linear relationships in the data.

Evaluation Metrics: Each model was evaluated using RMSE (Root Mean Squared Error) and R² Score. Lower RMSE indicates better prediction accuracy; higher R² indicates better variance explained by the model.

3. Results — Model Performance Comparison

Model	RMSE	R ² Score	Rank
Linear Regression	0.7456	0.5758	3rd
Ridge Regression	0.7455	0.5758	2nd
Decision Tree Regressor	0.7242	0.5997	1st ✓

4. Conclusion & Model Selection

The **Decision Tree Regressor** was selected as the best-performing model based on its superior metrics across both evaluation criteria. It achieved an RMSE of **0.7242** and an R^2 Score of **0.5997**, outperforming both Linear and Ridge Regression.

The key reason for this outcome is that housing price data contains non-linear relationships — for example, the relationship between location (latitude/longitude) and price is not linear. The Decision Tree naturally captures these patterns through recursive feature splits, while linear models assume a straight-line relationship between features and target.

Ridge Regression and Linear Regression performed nearly identically, confirming that L2 regularization provided minimal benefit on this dataset — likely because the features were already well-scaled and multicollinearity was not a significant issue.

Final Selected Model: Decision Tree Regressor (max_depth=5) | **RMSE:** 0.7242 | **R^2 :** 0.5997
